

10. Prove that there is a positive integer that equals the sum of the positive integers not exceeding it. Is your proof constructive or nonconstructive?

Solution: 3 is such a positive integer. The positive integers not exceeding 3 are 1 and 2. $1+2=3$. It's easy to see that my proof is constructive.

12. Prove that either $2 \cdot 10^{500} + 15$ or $2 \cdot 10^{500} + 16$ is not a perfect square. Is your proof constructive or nonconstructive?

Solution: Suppose, for the sake of contradiction, let both $2 \cdot 10^{500} + 15$ and $2 \cdot 10^{500} + 16$ be perfect squares. $\therefore$ By defn, $\exists$ integers m and n such that $2 \cdot 10^{500} + 15 = m^2$ and $2 \cdot 10^{500} + 16 = n^2$. $\therefore n^2 - m^2 = 1 \Rightarrow (n-m)(n+m) = 1$
 $\therefore n-m$ and $n+m$ are both integers, this is only possible for the following two cases:

Case 1: $n-m=1, n+m=1 \therefore 2n=2 \Rightarrow n=1. \quad 1-m=1 \Rightarrow m=0$
 $n^2=1 \neq 2 \cdot 10^{500} + 16 \quad m^2=0 \neq 2 \cdot 10^{500} + 15$

Case 2: $n-m=-1, n+m=-1 \therefore 2n=-2 \Rightarrow n=-1. \quad -1-m=-1 \Rightarrow m=0$
 $n^2=1 \neq 2 \cdot 10^{500} + 16 \quad m^2=0 \neq 2 \cdot 10^{500} + 15$

For both the cases we arrive at a contradiction.

$\therefore$ At least one of $2 \cdot 10^{500} + 15$ or $2 \cdot 10^{500} + 16$ is not a perfect square. $\square$

13. Prove that there exists a pair of consecutive integers such that one of these integers is a perfect square and the other is a perfect cube.

Solution: 8 and 9 are such a pair of consecutive integers.

$8 = 2^3$ and $9 = 3^2$. $\square$

14. Show that the product of two of the numbers $65^{1000} - 8^{2001} + 3^{177}$, $79^{1212} - 9^{2399} + 2^{2001}$, and $24^{4493} - 5^{8192} + 7^{1777}$ is non-negative. Is your proof constructive or nonconstructive?